

DN: A Pre-Generation Based Interactive Narrative System with Full-Ready Multimodal Delivery

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Abstract

Interactive narrative systems increasingly combine language generation, state tracking, and visual scene synthesis, but this richer output scope can make player-facing latency a central bottleneck. This draft reports the efficiency component of DN, a pre-generation based interactive narrative system that targets full-ready multimodal delivery. On a 20-item formal benchmark, simulated 60-second player reading time reduces DN’s mean next-turn full-ready latency from 27.095 seconds to 5.299 seconds and its median latency from 24.571 seconds to 0.027 seconds, while preserving a strict success rate of 1.000. Visual quality, narrative quality, and ablation results will be integrated in subsequent revisions.

1 Introduction

Interactive narrative systems must balance rich multimodal generation with low user-perceived latency. Unlike single-turn text or image generators, a playable narrative system must maintain story state, expose actionable options, advance the plot after a user decision, and deliver the next piece of content in a form that the player can immediately continue from. When the next turn contains both text and a scene image, the relevant efficiency question is therefore not merely how quickly a model can produce an isolated image, but how quickly the system can make the complete next turn usable.

DN addresses this problem through a pre-generation based delivery mechanism. Instead of waiting until the user clicks the next action to begin all downstream work, DN uses the player’s reading interval to prepare likely next-turn content in the background. This design turns reading time into computation time and aims to reduce the latency that remains visible to the player. The present draft first stabilizes the efficiency evidence for this mechanism. Later revisions will extend the

evaluation to visual quality, narrative/text quality, and component-level ablations, so that the final paper can jointly answer whether DN is fast enough for interaction and whether its delivered content is good enough for narrative play.

The current efficiency results support three stage-level findings. First, strict full-ready delivery without dwell time remains expensive because the system must wait for both textual and visual readiness. Second, a realistic reading interval substantially changes the latency distribution: in the pregen60 setting, most samples are delivered nearly immediately after the next click. Third, comparisons against image-continuation references should be interpreted as scope-aware references rather than fully isomorphic system comparisons, because DN returns text plus image plus playable state while the baselines report image or workflow latency.

2 System Overview

DN is an interactive narrative generation system that combines structured story-state management, option generation, and scene-image delivery. At each turn, the system presents narrative text and choices to the player, receives the selected action, updates the story state, and prepares the next multimodal scene. The paper’s central efficiency hypothesis is that pre-generation can reduce user-visible delay by using the idle interval between content delivery and the player’s next action.

The pre-generation mechanism maintains candidate next-turn work in the background. During the player’s reading interval, DN prepares likely continuations and their associated scene images. When the player later selects an action, the system attempts to serve a strict full-ready result from the prepared content; if the requested content is not yet fully available, the measured path waits until both text and image are ready. This distinc-

tion is important for the evaluation: a response is not counted as successful merely because a placeholder or partial output exists.

3 Experimental Setup

We evaluate efficiency on a 20-item formal benchmark subset derived from the DN-quality benchmark. Each item defines a narrative theme and constraints that can be mapped either to DN’s full interactive pipeline or to image-generation baseline adapters. The evaluation uses the latest v15 composite20 artifacts as the source of truth and excludes earlier v14, fixed-v2, and trimmed-sample exploratory results from the main paper tables.

The primary comparison is not intended to claim that all systems are architecturally identical. Instead, it measures the latency of DN’s full-ready delivery and reports image-baseline latencies as reference points for the image-continuation subtask. This scope-aware framing is necessary because DN performs story-state update and textual continuation in addition to scene-image delivery, whereas StoryDiffusion, SDM-v2, and IC-LoRA are evaluated as next-turn image or image-workflow references.

4 Efficiency Evaluation

Evaluation goal. The efficiency evaluation asks how long a player must wait after issuing a next-turn action before the system can deliver content that is ready to continue. This question is central to playable narrative generation: even high-quality narrative content can break interaction if the user must wait for a long or unpredictable generation process. We therefore evaluate DN under a *full-ready* latency definition, where a response is considered ready only when both the textual continuation and the corresponding scene image are available.

Protocol. We evaluate DN under two settings. The first setting, *normal20*, is a zero-dwell stress setting: the second click is issued immediately, so the system receives little opportunity to benefit from player reading time. The second setting, *pregen60*, simulates a player spending 60 seconds reading the current turn before clicking the next action. This setting tests the intended use case of pre-generation, where the system uses the player’s reading interval to prepare likely next-turn content in the background.

For external references, we compare against StoryDiffusion (Zhou et al., 2024), SDM-v2, which is based on latent diffusion modeling (Rombach et al., 2022), and IC-LoRA (Huang et al., 2024). These baselines are image-continuation references rather than complete interactive narrative systems: they generate a next-turn image or image workflow output from DN-style prompts, but they do not generate the full textual continuation, update the game state, or return a complete playable state. Consequently, the comparison is deliberately conservative in the sense that DN is measured on a heavier output scope. We keep this distinction explicit throughout the analysis.

Metrics. We report success rate, mean latency, median latency, and 95th-percentile latency (P95). For DN, latency is measured as `full_ready_elapsed_s`. For the image baselines, latency is measured as next-turn image or workflow latency. We also report mechanism-specific signals for DN, including direct image rate, likely hit rate, and placeholder rate. These signals help determine whether pre-generation improves latency by making the next-turn scene image available before the user clicks.

Main results. Table 1 shows that DN’s zero-dwell full-ready latency is 27.095 seconds on average, with a P95 latency of 46.196 seconds. This setting is intentionally strict: it measures the cost of producing a full next-turn response when the system is given essentially no reading interval in which to complete pre-generation. Under the simulated reading setting, DNpregen60 reduces the mean full-ready latency to 5.299 seconds and the median latency to 0.027 seconds. This substantial reduction indicates that pre-generation changes the distribution of user-perceived latency rather than merely improving a small number of cases.

Compared with the image-continuation references, DNpregen60 is faster on mean latency than StoryDiffusion (8.495 seconds) and IC-LoRA (30.057 seconds), while SDM-v2 remains faster at 2.827 seconds. This result should be interpreted with the output-scope distinction in mind: DN returns a full playable response that includes both text and the corresponding scene image, whereas the image baselines report only image or workflow continuation latency. Thus, the result supports the claim that DN’s pre-generation mechanism can make full-ready interactive delivery competitive with specialized image-generation

System	Output Scope	N	Mean (s)	Median (s)	P95 (s)	Success
DN normal20	Text + image full-ready	20	27.095	24.571	46.196	1.000
DN pregen60	Text + image full-ready	20	5.299	0.027	27.150	1.000
StoryDiffusion	Continuation image	20	8.495	–	8.900	1.000
SDM-v2	Continuation image	20	2.827	–	2.831	1.000
IC-LoRA	Continuation workflow	20	30.057	–	30.059	1.000

Table 1: Efficiency comparison on the formal20 benchmark. DNreports full-ready latency, where both the textual continuation and the corresponding scene image are available. Image baselines report next-turn image or workflow latency only.

DN Setting	Mean	Median	P95	Direct Image Rate
normal20	27.095	24.571	46.196	1.000
pregen60	5.299	0.027	27.150	1.000

Table 2: Effect of simulated reading time on DNfull-ready latency. The pregen60 setting simulates a 60-second dwell interval before the next click.

references under realistic player dwell time.

Effect of pre-generation. Table 2 isolates the mechanism-level effect of the simulated reading interval. Moving from normal20 to pregen60 reduces the mean full-ready latency from 27.095 seconds to 5.299 seconds and the median from 24.571 seconds to 0.027 seconds. The direct image rate remains 1.000 in both settings, and the placeholder rate is 0.0 in the source summary, indicating that the measured outputs are strict full-ready results rather than placeholder fallbacks. The likely-hit rate rises from 0.0 in normal20 to 0.8 in pregen60, which is consistent with the intended behavior of using reading time to prepare likely next-turn content.

Interpretation and limitations. The efficiency results provide a stable first piece of evidence for DN’s practical interaction profile. They show that the system is not merely generating richer multimodal content at arbitrary latency: when pre-generation has a realistic opportunity to operate, full-ready delivery becomes competitive with several image-only references. At the same time, the comparison is not a fully isomorphic system-to-system comparison. StoryDiffusion, SDM-v2, and IC-LoRA do not perform DN’s state update and textual continuation steps. We therefore use them as image-continuation efficiency references, while reserving separate sections for visual quality and narrative quality evaluation.

Stage-level conclusion. The current efficiency results are ready to support the first experimental subsection of the paper. They establish that DN’s pre-generation mechanism substantially improves next-turn delivery latency, reducing average full-ready latency by 80.4% relative to the zero-dwell setting. The remaining quality question is orthogonal: the next sections will evaluate whether the content delivered quickly is also visually faithful, narratively coherent, and consistent across turns.

5 Visual Quality Evaluation

This section evaluates whether the images delivered by each system are usable for interactive narrative play. While Section 4 measures how quickly a system can deliver the next playable state, visual quality measures whether the delivered image follows the requested narrative theme and whether the visual world remains aesthetically stable across turns.

Protocol. We evaluate the same formal20 benchmark used in the efficiency study. For DN, we use the pre-generation full-ready outputs from the main setting. For StoryDiffusion, SDM-v2, and IC-LoRA, we use their existing formal20 first-turn and next-turn image outputs. No image is regenerated for this evaluation. Each system contributes three image records per benchmark item: the first-turn image, the current image used for the next-turn continuation, and the next-turn image. This yields 60 image-level records per system and 240 image-level records in total.

Instruction following and theme alignment. For each image, a vision-language judge scores whether the visual output follows the benchmark theme, the current scene text, the requested image style, and visually checkable must-have constraints. The judge also records forbidden-issue violations such as visible watermarks, unintended

text, style drift, or theme leakage. The primary holistic score is *Instruction Following*, reported on a 1–5 scale.

Aesthetic consistency. For each system–benchmark-item group, the judge scores the three associated images jointly. The rubric covers style and lighting consistency, subject-attribute consistency, scene-world consistency, composition quality, and artifact severity. The primary holistic score is *Aesthetic Consistency*, also reported on a 1–5 scale. This group-level evaluation is intentionally distinct from single-image appeal: it asks whether the same system preserves a coherent visual identity as the narrative advances.

Current status. The evaluation manifest has been built from existing formal20 artifacts. Image validation found one invalid zero-byte SDM-v2 first-turn image; all other image records are readable. Table 3 reports a reproducible automatic proxy based on CLIP image–text similarity, zero-shot style ranking, cross-turn image-embedding consistency, color-histogram consistency, and low-level artifact checks. Because this proxy does not replace direct semantic inspection, the final version should confirm these numbers with VLM or human judges.

6 Narrative and Text Quality Evaluation

This section is reserved for narrative and text-quality evaluation. Planned metrics include world-view coherence, option actionability, plot continuity, state consistency, and human or model-judge ratings. These experiments will determine whether DN’s textual and structured narrative outputs remain coherent and playable beyond the latency measurements reported in Section 4.

7 Ablation and Mechanism Analysis

The current main-ready mechanism analysis focuses on pre-generation, because this is the only ablation-style result that is available under the latest v15 composite20 protocol used by the main efficiency table. We therefore treat the normal20 and pregen60 settings as a controlled mechanism comparison over the same 20 benchmark items: normal20 measures strict next-turn full-ready latency with zero dwell time, while pregen60 keeps the same benchmark scope but gives the system a simulated 60-second player reading interval in which likely next-turn content can be prepared.

Table 4 shows that the simulated reading interval substantially changes both the central tendency and the hit behavior of DN. The mean full-ready latency decreases from 27.095 seconds to 5.299 seconds, and the median decreases from 24.571 seconds to 0.027 seconds. At the same time, the likely-hit rate increases from 0.000 to 0.800, indicating that the system often has the requested next-turn content ready by the time the user clicks. This result supports the mechanism-level interpretation behind the efficiency results in Section 4: the latency gain is not only a change in reporting protocol, but is associated with successful use of the reading interval for background preparation.

Other mechanism studies are not yet included as main ablation results. Existing read-wait threshold runs and council/no-council comparisons are useful internal evidence, but they were produced under earlier benchmark versions or different sample sizes. We therefore reserve them for either an appendix clearly marked as exploratory or a rerun under the same formal20/v15 composite20 protocol before they are used as main-paper ablations.

8 Conclusion

The final conclusion will be completed after the quality-evaluation sections are integrated. The current efficiency results already show that pre-generation substantially reduces DN’s full-ready latency under a realistic reading-time simulation.

References

- Lianghua Huang, Wei Zhang, Yuxin Chen, Yu Wang, Zhe Jin, Xuan Wu, Wen Jiang, and Ziwei Liu. 2024. [Ic-lora: In-context learning of lora for diffusion transformers](#). *arXiv preprint arXiv:2410.23775*.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. 2022. [High-resolution image synthesis with latent diffusion models](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.
- Yupeng Zhou, Daquan Zhou, Ming-Ming Cheng, Jiashi Feng, and Qibin Hou. 2024. [Storydiffusion: Consistent self-attention for long-range image and video generation](#). *arXiv preprint arXiv:2405.01434*.

A Experiment Artifact Provenance

The efficiency tables in Section 4 use the latest v15 composite20 artifacts. Earlier exploratory runs, including v14, fixed-v2, and trimmed 16-sample

System	N_{img}	N_{grp}	Instruction Following	Aesthetic Consistency	Theme Violation Rate
DN	60	20	4.386	4.619	0.033
StoryDiffusion	60	20	3.620	4.485	0.017
SDM-v2	59	20	3.008	4.764	0.136
IC-LoRA	60	20	2.351	4.403	0.467

Table 3: Automatic CLIP+CV proxy visual-quality summary on the formal20 benchmark. Instruction Following is scored per image; Aesthetic Consistency is scored per system–benchmark group. Scores range from 1 to 5. SDM-v2 has one invalid zero-byte first-turn image, so its image-level count is 59. These automatic proxy scores should be replaced or confirmed by VLM/human judging before final submission.

Setting	N	Mean	Median	P95	Likely Hit
normal20	20	27.095	24.571	46.196	0.000
pregen60	20	5.299	0.027	27.150	0.800

Table 4: Main-ready mechanism comparison for pre-generation. Both rows use the latest v15 composite20 full-ready protocol. Latencies are measured in seconds.

Component	Current Status	Main-Text Use
Pre-generation	v15 composite20, $N = 20$	Main-ready
Read-wait thresholds	v10–v13 exploratory runs	Appendix or rerun
Council planning	benchmark-v1 exploratory runs	Appendix or rerun
Quality-side effects	Not yet human-validated	Reserved

Table 5: Status of available ablation evidence. Only the pre-generation comparison currently matches the latest main efficiency protocol.

variants, are retained only for internal analysis and are not used in the main paper numbers.

B Per-Sample Efficiency Details

A future appendix revision will include per-sample latency rows from the v15 composite20 table to support sensitivity analysis and long-tail inspection.

Artifact	Role in Paper	Path
strict full-ready comparison	Main cross-system efficiency table	experiments/handoff.2026-04-30.main.experiments/01.main.tables/strict_fullready_main_comparison.2026-05-01.v6.composite20.csv
DN normal vs. pregen60	DN mechanism table	experiments/handoff.2026-04-30.main.experiments/01.main.tables/dn_fullready_normal_vs.pregen60.2026-05-01.v4.composite20.csv
v15 full-ready summary	Summary validation and mechanism signals	experiments/handoff.2026-04-30.main.experiments/01.main.tables/benchmark.v15.fullready.nextturn.20.summary.composite20.v1.json
per-sample v15 table	Per-item appendix candidate	experiments/benchmark/standard_runs/benchmark.v15.fullready.nextturn.20.table.composite20.v1.csv

Table 6: Source-of-truth artifacts for the efficiency evaluation. Paths are repository-relative.